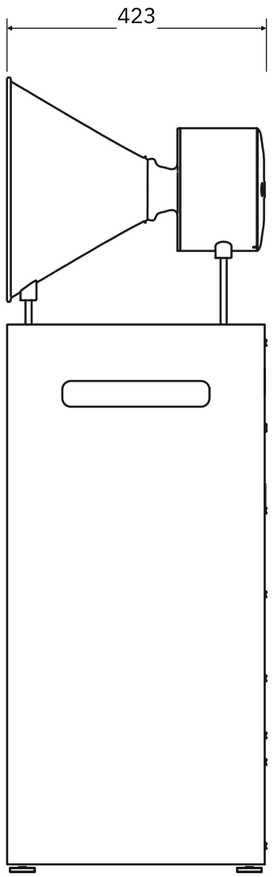
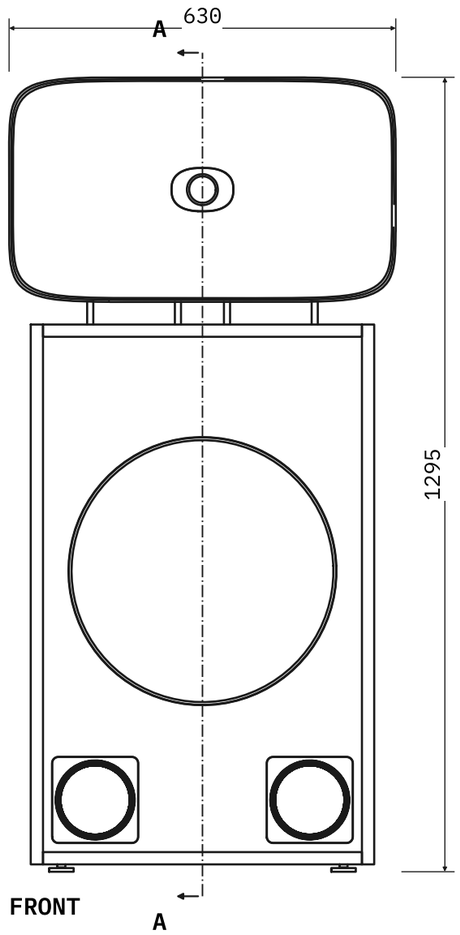
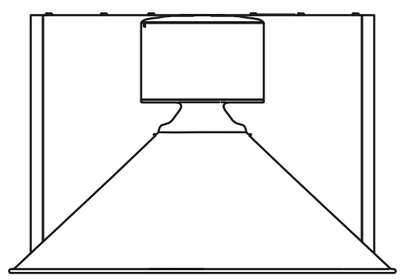
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

General arrangement

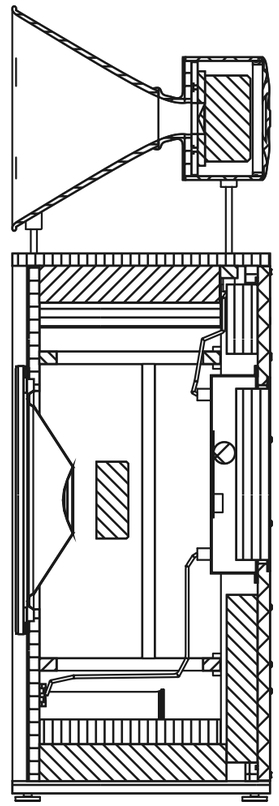
barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

BHC-001 rev A

Scale 1:10. Section A-A (sheet 1): the product cut at x 280.0 and seen from the right; cut faces hatched, a different angle for each part.



Section A-A

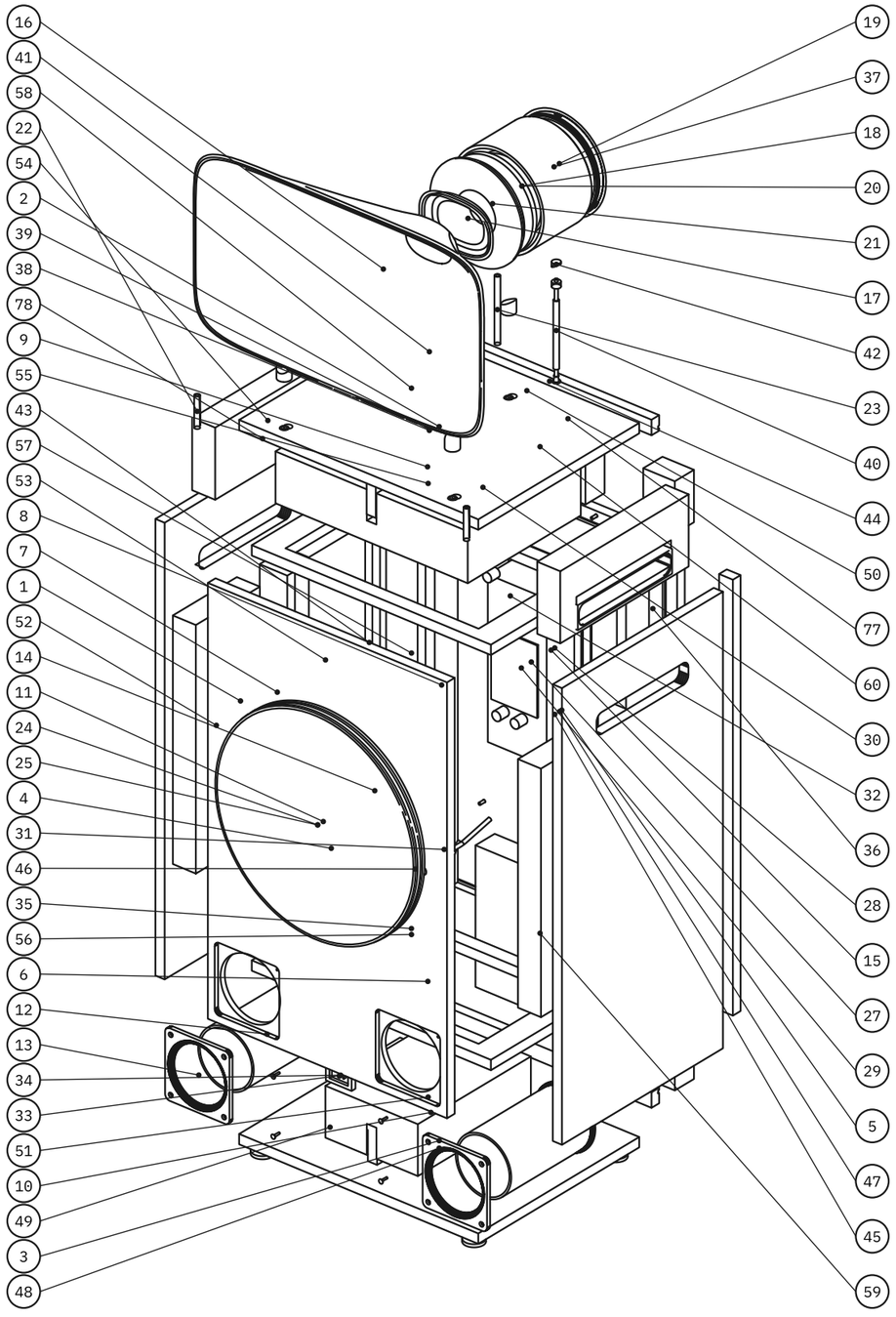
barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

BHC-002 rev A

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).



Exploded view

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

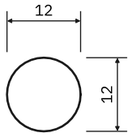
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

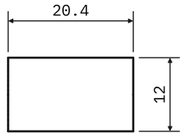
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

79. ir-window: lathe, clear-cast-acrylic; 2 cm3 each; 1.0 for the set; 1:1

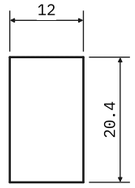
only if the 940 nm test fails: a clear cast acrylic rod 12 across, the panel bored through in front of the receiver's eye and the rod bonded in flush, its face polished; a clear dot in the tinted panel



FRONT



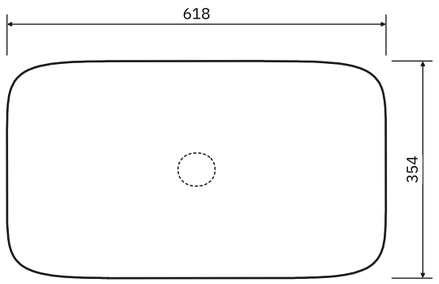
FROM THE RIGHT



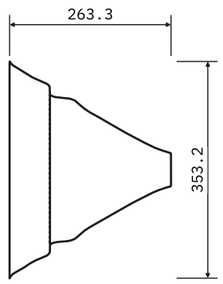
FROM ABOVE

80. horn-tool-prototype: cnc-mill, tooling-board; 19015 cm3 each; 1.0 for the set; 1:10

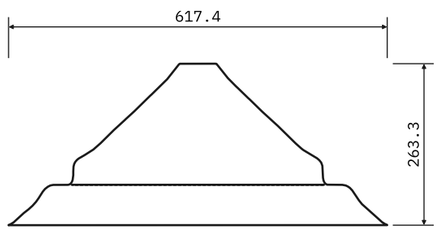
the prototype's plug (the coordinator's call, round 36): the production plug's surface cut by the same 5-axis program from a glued-up block of high-temperature epoxy tooling board (rated 150 C or more), sealed and polished, vented as the aluminium; on the production base plate and trim fixture, which carry over. It forms the same 10 mm sheet at 4 bar over the same male geometry, so the prototype tests the production process, material and profile; its face is the air path, which sealed board holds as well as aluminium for a short run (some tens of shots), and the outside is the free ...



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

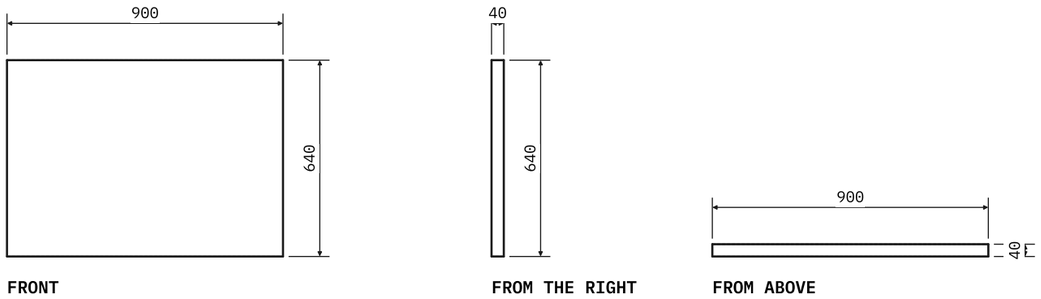
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

81. horn-tool-base: cnc-mill, aluminium; 23040 cm3 each; 1.0 for the set; 1:20

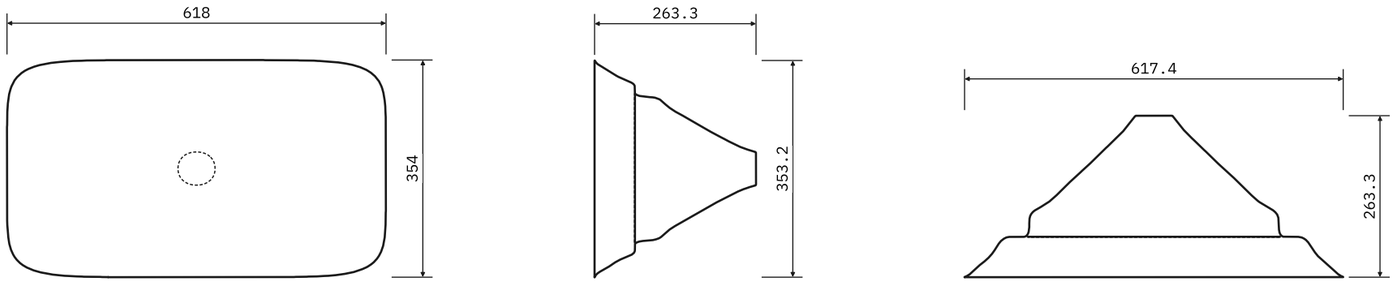
the tool's base plate in the mouth's plane, 6061 machined, the clamp frame's seal groove round it and the plug bolted on from below, so the flange forms flat on it; one tool makes every horn (the line's quantity is one)



FRONT FROM THE RIGHT FROM ABOVE

82. horn-tool: cast, aluminium; 19015 cm3 each; 1.0 for the set; 1:10

the forming tool's plug: the horn's air path from its nose 25 behind the trim (its edge rounded) to the mouth, its lip a cove R12 at its foot; A356 aluminium cast near shape from a pattern and machined all over on a 5-axis mill to the air path's surface, polished; drilled for oil at 85 C, vented by 0.6 mm holes in the waste and the mouth's corners; one tool makes every horn (the line's quantity is one)



FRONT FROM THE RIGHT FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. The waveguide stands on four posts, stainless lamp tube 10 × 7, each threaded M10 × 1 at its top into a brass insert bonded in a pad cast under the horn (drilled after casting, through a jig taken off the top's sockets), its plain foot standing in a polyurethane sleeve in a brass socket bonded with epoxy into the top, so the horn and its posts lift off as one. An M6 rod down the free rear post, from a thumb nut inside the pod to an insert under the top, holds them down. The cap closing the pod is held by a twist-lock (round 16): three lugs on its spigot in L-slots in the pod's bore, locked by an eighth of a turn, an O-ring beyond them; no screw shows. The HF lead, a round silicone cable 5 across, runs from its connector at the amplifier's compartment straight up, through a gland under the top and up inside a rear post into the pod, where it plugs into the driver's leads. To lift the horn off: the cap off, the nut undone and the lead unplugged.
2. The FA122 crosses at 1 kHz, LR4 (round 35: in its simulation the horn loads the driver fully only from about 900 Hz, and two-way speakers of this size cross where their horns allow: JBL's 4367 at 700 Hz and its M2 at 800 on deeper waveguides, its 4430 at 1 kHz; 800 to 900 Hz only if the first sample's distortion at 100 dB/1 m stays low; B&C asks 12 dB/oct or steeper, at 800 Hz or above) with the HF channel about 6 dB down (105 dB at 2.83 V against the woofer's 99), part of it in the passive L-pad after the amplifier, so its hiss stays inaudible: two 25 W aluminium-housed resistors on a plate in the compartment, behind a 33 uF film capacitor that keeps low frequencies off the driver if the preset is ever lost.
3. The amplifier's preset, stored before the speaker ships: the crossover; a 4th-order Butterworth high-pass at 28 Hz on the woofer (below the ports' tuning the box no longer holds the cone, which would pass Xmax below about 30 Hz at full output, and a record's warp drives it there); limiters set to the woofer's Xmax and the driver's rating; a notch at the ports' pipe resonance (about 540 Hz in the pressed cabinet, 560 in the Cloud's, 600 in the plywood one), measured in each port's near field; and Hypex's auto-standby on (round 35: about 0.5 W asleep, and the compartment cooler in a warm room).
4. The amplifier in a sealed compartment of its own, as Hypex asks: 1.5 mm aluminium, black anodised, on an EPDM gasket on the back's inner face, so the box stays airtight and the mains electronics are enclosed in metal. The leads leave it through cable glands. A rating plate beside the amplifier carries the model, the serial number, the ratings, the marks and the maker's address; the finished speaker needs an EMC pre-scan and an IEC 62368-1 assessment.
5. The lining: polyester batts (thermally bonded, recycled PET), 50 mm on the sides and 100 mm on the top and the bottom (round 21), one to each bay between the braces (and the ribs), and 50 mm on the back, its full height between its ribs, so they come off with it; each cut 10 full each way and sewn into a sleeve of white voile; none on the front panel, none beside the ports (10 clear of each tube, nothing between its inner end and the back's batts) and none within 20 of the amplifier's compartment.
6. The back comes off (round 14): it closes on a 3 mm EPDM gasket against four land strips inside the shell, sixteen M5 button heads through it into inserts in them. The amplifier's compartment, the back's ribs and its batts come off with it and the leads unplug at the compartment, so the amplifier, the L-pad and the leads can be reached.
7. Four levelling feet (M8, 40 swivel bases on rubber pads) in inserts in the bottom, 50 in from its edges, lift the cabinet 12 mm: set them so it stands level and cannot rock under the horn, and lock their nuts.
8. No wood: every panel is cast acrylic, 20 mm, tinted through in its colour (red, orange, amber, green, blue, violet, frosted, black), recycled (made from recycled MMA) where a maker can make the colour, cut on a CNC router from 4 × 8 ft sheets of one batch (approve a tint swatch and a cemented corner sample first). The ring (6 mm) is cast acrylic tinted opaque in the ring colour; the cloth disc's frame (12 mm) black cast acrylic.
9. Assembly: each panel's edges milled square and polished where they show, the panels annealed (about 80 C, an hour for each 3 mm of thickness, cooled slowly); the ribs cemented to their panels flat (the back's to the back); the shell's corners, the front panel (set back 24 mm), the two braces and the back's four land strips cemented with a two-part polymerising acrylic cement for thick sheet (ACRIFIX 2R 0190 or Weld-On 40), clamped square on a jig, no visible bubble in any joint; the box annealed again; the lands' faces, the bosses' and the seal's groove milled in one setup; the back closes the box last, seated hard on the lands and the braces' bosses over a cord seal, twenty-two screws, and comes off. No solvent cement: it crazes thick cast sheet.

Notes

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

10. The woofer flush in its rebate on a gasket, on eight M6 brass inserts bonded with epoxy in 8.5 holes; the ring over its frame on its magnets, bonded in their pockets; each port in the cabinet's own acrylic, its tube cemented into its 8 mm frame, fitted from the front into a recess on a foam gasket, four black M4 screws in its corners into tapped threads; the tubes cut to the length found on the first sample before their ends are flared.
11. The front: the panel's own tinted acrylic, the lining showing through it round the woofer, and a cloth disc over the woofer in the acrylic's colour, its frame standing on the ring on four magnet pairs.
12. Seen through: everything inside is drawn to be seen. The leads run straight and tied, the glands and the brass sockets show, the amplifier's compartment is black anodised, and each batt's voile sleeve has its seams to the wall.
13. Lifting (round 24): a grip 240 × 42 for both hands through each side, 92 below the top, a cup of 3 mm acrylic cemented behind it and the side's batt cut round it; two people lift a speaker by its grips, the horn off. The feet's M8 inserts take castors, spikes or isolation feet in place of the levelling feet, sold separately.
14. The mark (round 24): EARF 140 wide, laser-etched high on the back, centred, the etch filled with the speaker's own colour, deeper.

Notes

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